

CSE4356/CSE5392/EE5315/EE4328 (Fall 2025)

Lab #6

In this lab, you will add the loadable kernel module that controls the MCP23008 I2C expander using a kernel module.

The following steps will guide you through this process:

1. Write a kernel module (/sys/class/i2c or similar) to control the I2C IP module. Adapt your C code or use a bash script to leverage the sysfs interface. You should support the following files and values:

File	Write value	Read value
/sys/kernel/i2c/mode	read or write	last written value
/sys/kernel/i2c/byte_count	0-15	last written value
/sys/kernel/i2c/register	0-255 or none	last written value
/sys/kernel/i2c/address	0-127	last written value
/sys/kernel/i2c/use_repeated_start	true or false	last written value
/sys/kernel/i2c/start	any value	current status
/sys/kernel/i2c/tx_data	value for TX_FIFO	N/A
/sys/kernel/i2c/rx_data	N/A	value in FIFO or -1 if empty

2. 7 Bus Expander Control (CSE 5392, EE 5315, or Honors Contract)

Write a kernel module (/sys/class/i2c_expander) to control the I2C IP module, abstracting the functions of the I2C expander. Adapt C code or use a bash script to access this interface and display I2C expander information. You should support the following files and values at a minimum:

File	Write value	Read value
/sys/kernel/i2c_expander/pin/dir	direction of pin	last written value
/sys/kernel/i2c_expander/pin/pullup	pull up control on pin	last written value
/sys/kernel/i2c_expander/pin/data	data on port output	data on port pin

where /pin/ is the pin number (0-7 for MCP23008)

3. Connect SDA and SCL to the MCP23008 IC. Pullups should be connected to both outputs. At least one output and one input must be connected to the MCP23008 to show operation. Write a program similar to the gpio utility and stop go program to test operation.

4. Please implement all features of the project and submit by the due date in the syllabus.

6. Send the i2c.v file and kernel and user space C programs to the TA.

7. You should review the project to verify that all other features are implemented.

This lab is due by December 2, 2025.